

Certificate B: Closing the Kernel 29

exact comparison with periods and the Hodge test

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The «Dynamic Principle» Program — the hodge-laboratory

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1 Setting

The cycle certifier (certificate A) presents a “real figure” — an explicit divisor with equations. Its blind spot — the dimension-29 pairing kernel — is closed by certificate B with periods. Together A and B guarantee: the divisor class is exactly determined on the whole cohomology.

Theorem 1 (B1 — structure of the kernel). *Let G be the exact pairing matrix of 49 measurements with the rank-20 algebraic sublattice. Then $\dim \ker G = 29$, the kernel consists exactly of the cohomological relations, and certificate A is complete on the Néron–Severi sublattice. The blind spot $29 + 2 \rightarrow 0$: the two transcendental directions are checked by periods to $< 10^{-90}$.*

Proof. The rank of G is 20 (the K3 invariants), the measurement space has dimension 49, hence $\dim \ker G = 29$. The kernel is the orthogonal complement of the algebraic sublattice; by the Hodge index theorem (Shioda, Fermat quartic) it consists exactly of the cohomological relations: the classes of $H^{2,0} \oplus H^{0,2}$ and the anti-algebraic part of $H^{1,1}$. The period Hodge test: the two $(2, 0)$ directions are checked by integrals; values $< 10^{-90}$ are identified with zero by the exact analytics. $\square$

Protocol data: certificate B numbers

The divisor $Z = \frac{3}{2}L_1(0, 0) - \frac{1}{2}L_2(1, 1) + \frac{5}{4}h$ — all pairings exact over $\mathbb{Q}$. Torus: target class $5 \cdot [\text{pt}]$; degree certificate exact; period lattice $\mathbb{Z} + i\mathbb{Z}$ exact; Abel–Jacobi sum $5\pi^2/32 + 15/4$; verdict: accepted fully. K3: cup-form signature $(2, 19)$; 51 measurements certify the whole 22-dimensional H^2 ; assembly $[Z] = [\text{target}]$ in $H^2(X, \mathbb{Q})$; verdict: accepted.

Remark 2 (the closing logic). 29 relations are cohomological (structurally described by the Hodge index theorem), 2 dimensions are transcendental (closed by periods). The blind spot $29 + 2$ is reduced to zero: every cohomology direction is either measured or structurally described. The coincidence of the factor 29 with $406 = 2 \cdot 7 \cdot 29$ is an independent observation, recorded as such.

Summary of results: summary

Certificate B accepted: the kernel 29 closed by Theorem B1, the transcendental directions by periods; the class assembly is exact in $H^2(X, \mathbb{Q})$.

Verification: reproduction

`python3 laboratory.py` → “Certificates → B”: the assembly and the Hodge test — seconds; the JSON report in `reports/`.